

Safe Harbor 2.0

Some people may argue Safe Harbor 2.0 is the best option, but Max Schrems explains why it's another doomed attempt at connectivity between the continents. <http://digit.in/LostSafe>

**Sync your 'fitness'**

Oklahoma's Oral Roberts University has made it mandatory to own a Fitbit for a course which counts for 20% of grades. <http://digit.in/SyncFit>



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Hello world for sensors with Arduino!

You've obviously heard of the Arduino boards and how they've revolutionised hobby electronics. Here's a little primer to get you started.



Praneet Sab

You've heard a lot about 'Internet of Things' and probably bought yourself an Arduino kit along with a few more sensors but you can't figure out how to exactly use them! If that sounds like you, then you're at the right place.

Before we begin, I'll recommend you to skim through the "Fast Track to Arduino for Everyone" (Available with Digit's December, 2014

issue) and of course, if you want to see a preview of what you can expect to build after reading this article, then open up the December 2015 issue and check out the DIY article about Home Automation (pg. 132). And of course, I'm assuming that you know basic programming stuff, knowing how 'if' statements work & what libraries are, is the bare minimum for this article.

Here's what you'll need:

- An Arduino Uno (Obviously!)
- Jumper wires (Make it yourself!)
- Breadboard (Optional, but helpful)
- Arduino IDE (Get it here <http://bit.ly/1R2xniM>)

There are undoubtedly a lot of sensors available and covering all of them won't be feasible and that's why I'll cover the most frequently used sensors. So let's get started!

Ultrasonic Distance Sensor

You would have probably read about the SONAR that's used in ships. By taking the same concept into a small form factor we get a distance sensor. This sensor is available in multiple variants and I'll be using the 'HC-SR05' variant. It can easily detect range between 2 to 450 cm and has 5 different pins, out of which, we don't need the 'OUT' pin for now. To brief you a little, the 'trigger' pin will send the pulse and the 'ECHO' pin would wait for it to return. The other two pins, GND & VCC, are common and found on almost every component, 'GND' pin stands for 'ground' and the 'VCC' pin means '5v input'.

The connection is fairly simple; you need to connect the sensor to the Arduino directly (you may also use a breadboard). Using the jumper wires, connect the 'GND' pin to the 'GND' port of the Arduino and connect the 'VCC' pin to the '5v' port on your Arduino. The 'TRIG' and 'ECHO' pin can be connected to any

of the Arduino's Digital pins. For now, I'm connecting 'TRIG' to Digital pin no. 12 and the 'ECHO' to pin no. 11. And don't forget to connect the Arduino to your computer using the USB cable.

The beauty of these sensors is that most of them have a library function available, which basically ease our work since we don't need to re-invent the wheel. The library for this sensor is named as "NewPing" and you can get it from <http://bit.ly/1pYs3OV> and then unzip it into the Documents → Arduino → libraries folder. Open up the Arduino IDE (or restart in case it was already open) and on the menu bar, navigate to File → Sketchbook → libraries → NewPing → NewPingExample. Clicking on it would open up another Arduino window that has got all the code necessary to use the Distance Sensor. Click on the Upload button and a few LEDs would blink on the Arduino. After 2-3 seconds open up the "Serial Monitor" under 'Tools' menu option and there you go! You'd be seeing the distance between the sensor and the object in front of it. Using a flat surfaced object will get you better results.

You don't happen to write any code this time since most of it is handled by the library. However, if you've noticed then the distance is actually returned by the function named 'sonar.ping_cm()'. So, yeah, go on, assign that to a variable and use the variable for your programs. You can make a car that turns automatically if something comes within 'x' units or maybe, automate the lights in your home, so that if something comes within 'x' units then turn on or off the lights.

